B3: Quantum, Atomic and Molecular Physics — Model Solutions, 2018

Question 1 — Fine structure of hydrogenic systems

(a) Origin of fine structure and derivation of ΔE_{so}

Problem. Explain the physical origin of fine structure in hydrogenic atoms; derive $\Delta E_{\mathrm{so}} = \frac{\beta}{2}[j(j+1) - l(l+1) - s(s+1)]$ and find β.

Physical origin: in the electron's rest frame the proton orbits, generating a magnetic field $\mathbf{B}$; the electron spin magnetic moment $\boldsymbol{\mu}_s = -g_s\mu_B\mathbf{s}/\hbar$ couples to $\mathbf{B}$, giving the spin–orbit interaction (a relativistic correction $\sim \alpha^2$).

Spin–orbit Hamiltonian (Thomas-corrected):

$$H_{\mathrm{so}} = \frac{1}{2m_e^2c^2}\,\frac{1}{r}\frac{\mathrm{d}V}{\mathrm{d}r}\,\mathbf{L}\cdot\mathbf{S}, \qquad V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}.$$

Differentiate V:

$$\frac{1}{r}\frac{\mathrm{d}V}{\mathrm{d}r} = \frac{Ze^2}{4\pi\epsilon_0}\,\frac{1}{r^3}.$$

Define $\xi(r) = (2m_e^2c^2)^{-1}(1/r)(\mathrm{d}V/\mathrm{d}r)$:

$$H_{\mathrm{so}} = \xi(r)\,\mathbf{L}\cdot\mathbf{S}.$$

Use $\mathbf{J} = \mathbf{L} + \mathbf{S}$ to eliminate $\mathbf{L}\cdot\mathbf{S}$:

$$\mathbf{J}^2 = \mathbf{L}^2 + \mathbf{S}^2 + 2\mathbf{L}\cdot\mathbf{S}.$$

Solve for $\mathbf{L}\cdot\mathbf{S}$:

$$\mathbf{L}\cdot\mathbf{S} = \tfrac{1}{2}\big(\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2\big).$$

Eigenvalues in $|n\,l\,s\,j\,m_j\rangle$:

$$\langle \mathbf{L}\cdot\mathbf{S}\rangle = \tfrac{\hbar^2}{2}\big[j(j+1) - l(l+1) - s(s+1)\big].$$

Take the radial expectation $\langle\xi(r)\rangle$:

$$\Delta E_{\mathrm{so}} = \langle\xi\rangle\,\tfrac{\hbar^2}{2}\big[j(j+1) - l(l+1) - s(s+1)\big].$$

Hence $\beta = \hbar^2\langle\xi(r)\rangle$. Insert $\langle 1/r^3\rangle$ from the hint:

$$\beta = \frac{\hbar^2}{2m_e^2c^2}\,\frac{Ze^2}{4\pi\epsilon_0}\Big(\frac{Z}{na_0}\Big)^3\frac{1}{l(l+\frac{1}{2})(l+1)}.$$

Use $a_0 = 4\pi\epsilon_0\hbar^2/(m_e e^2)$ and $\alpha = e^2/(4\pi\epsilon_0\hbar c)$:

$$\boxed{\beta = \frac{Z^4\alpha^2}{n^3 l(l+\frac{1}{2})(l+1)}\,\frac{|E_n|}{1}\cdot\frac{1}{1} = m_ec^2\,\frac{Z^4\alpha^4}{2n^3 l(l+\frac{1}{2})(l+1)}}$$

(equivalently $\beta = -E_n\,Z^2\alpha^2/[nl(l+\frac{1}{2})(l+1)]$ with $E_n = -\frac{1}{2}m_ec^2Z^2\alpha^2/n^2$).

(b) Fine structure of $n = 2, 3$ in He^+

He^+: $Z = 2$, $s = 1/2$. Fine-structure constant prefactor:

$$E_h = m_e c^2 \alpha^2 / 2 = 27.211\,\mathrm{eV}.$$

General level shift in units of $m_e c^2 \alpha^4$:

$$\Delta E_{\mathrm{so}} = \frac{Z^4}{2n^3} \frac{j(j+1) - l(l+1) - s(s+1)}{l(l+\frac{1}{2})(l+1)} m_e c^2 \alpha^4.$$

Tabulate (with $Z^4 m_e c^2 \alpha^4 = 16 \times 0.7251\,\mathrm{meV} = 11.60\,\mathrm{meV}$ for the overall scale; $m_e c^2 \alpha^4/2 = 0.3625\,\mathrm{meV}$):

$n = 2$: levels $2s_{1/2}$, $2p_{1/2}$, $2p_{3/2}$.

- $2s_{1/2}$: $l = 0$ (no SO shift; included for completeness).
- $2p_{1/2}$: $j(j+1) - l(l+1) - s(s+1) = 3/4 - 2 - 3/4 = -2$; $l(l+\frac{1}{2})(l+1) = 3$.

$$\Delta E(2p_{1/2}) = \tfrac{Z^4}{2n^3} \cdot \tfrac{-2}{3} \cdot m_e c^2 \alpha^4 = -\tfrac{16}{16} \cdot \tfrac{2}{3} \cdot 0.3625\,\mathrm{meV} \cdot 2$$

Insert numbers: $\Delta E(2p_{1/2}) = -0.483\,\mathrm{meV}$.

- $2p_{3/2}$: factor $= 15/4 - 2 - 3/4 = 1$; shift $\Delta E(2p_{3/2}) = +0.242\,\mathrm{meV}$.

$n = 3$: levels $3s_{1/2}$, $3p_{1/2}$, $3p_{3/2}$, $3d_{3/2}$, $3d_{5/2}$.
For $n = 3$ the prefactor $Z^4/(2n^3) = 16/54$. Compute angular factors:

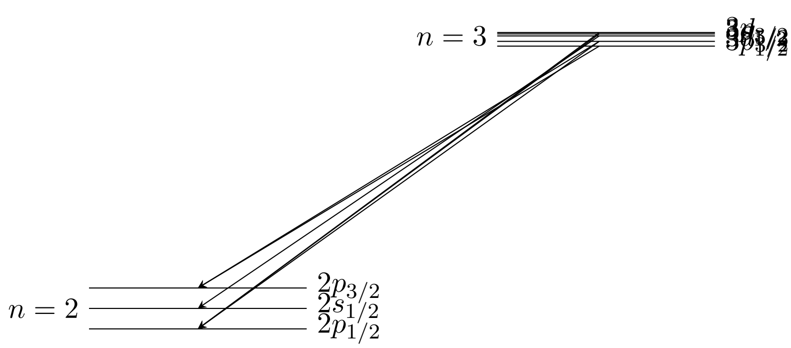
$$3p_{1/2}: \frac{-2}{3}, \quad 3p_{3/2}: \frac{1}{3}, \quad 3d_{3/2}: \frac{-3}{15/2} = \frac{-2}{5}, \quad 3d_{5/2}: \frac{2}{15/2} = \frac{4}{15}.$$

Multiply by $16/54 \cdot m_e c^2 \alpha^4 = 0.215\,\mathrm{meV}$:

$$\Delta E(3p_{1/2}) = -0.143\,\mathrm{meV},\ \Delta E(3p_{3/2}) = +0.0716\,\mathrm{meV},$$

$$\Delta E(3d_{3/2}) = -0.0859\,\mathrm{meV},\ \Delta E(3d_{5/2}) = +0.0573\,\mathrm{meV}.$$

Energy-level diagram:



Allowed transitions ($\Delta l = \pm 1$, $\Delta j = 0, \pm 1$, $j = 0 \not\to 0$):

$$3p_{1/2} \to 2s_{1/2},\ 3p_{3/2} \to 2s_{1/2},\ 3s_{1/2} \to 2p_{1/2},\ 3s_{1/2} \to 2p_{3/2},$$

$$3d_{3/2} \to 2p_{1/2},\ 3d_{3/2} \to 2p_{3/2},\ 3d_{5/2} \to 2p_{3/2}.$$

(c) Ω^- bound to bare Pb (Z=82), $n = 10$, $l = 9$

Splitting into 4 levels means $2s{+}1 = 4$, hence $\boxed{s_\Omega = 3/2}$ (so $j = l{-}s, \ldots, l{+}s = 15/2, 17/2, 19/2, 21/2$).

Reduced mass effect: replace $m_e \to \mu = m_\Omega m_{\text{Pb}}/(m_\Omega + m_{\text{Pb}}) \approx m_\Omega$ (Pb nucleus much heavier). The Bohr-like formula gives, with $m_\Omega c^2 = 1672\,\text{MeV}$:

$$\beta = \frac{Z^4 \alpha^4 m_\Omega c^2}{2n^3 l(l+\frac{1}{2})(l+1)}.$$

Insert $Z^4 = 82^4 = 4.5212 \times 10^7$, $\alpha^4 = (1/137.036)^4 = 2.834 \times 10^{-9}$:

$$Z^4\alpha^4 = 0.1281.$$

Insert $n^3 = 1000$, $l(l + \frac{1}{2})(l+1) = 9 \cdot 9.5 \cdot 10 = 855$:

$$\frac{Z^4\alpha^4}{2n^3 l(l+\frac{1}{2})(l+1)} = \frac{0.1281}{2 \cdot 1000 \cdot 855} = 7.49 \times 10^{-8}.$$

Multiply by $m_\Omega c^2 = 1.672 \times 10^9\,\text{eV}$:

$$\beta = 7.49 \times 10^{-8} \times 1.672 \times 10^9\,\text{eV} = 125.2\,\text{eV}.$$

Energy of level j:

$$E_j = \tfrac{\beta}{2}\big[j(j+1) - l(l+1) - s(s+1)\big] = \tfrac{\beta}{2}\big[j(j+1) - 90 - \tfrac{15}{4}\big].$$

Successive separations $E_{j+1} - E_j = \beta(j+1)$:

$$E_{17/2} - E_{15/2} = \tfrac{17}{2}\beta, \quad E_{19/2} - E_{17/2} = \tfrac{19}{2}\beta, \quad E_{21/2} - E_{19/2} = \tfrac{21}{2}\beta.$$

Insert $\beta = 125.2\,\text{eV}$:

$$\boxed{\Delta E_{15/2\to17/2} = 1.06\,\text{keV}, \;\; \Delta E_{17/2\to19/2} = 1.19\,\text{keV}, \;\; \Delta E_{19/2\to21/2} = 1.31\,\text{keV}.}$$

(d) Two Ω^- bound to Pb: configurations $1s2l$ and ground

The Ω^- has $s = 3/2$, hence is a fermion (half-integer spin). Pauli exclusion still applies between identical fermions: total wavefunction antisymmetric under exchange.

Excited $1s\,2l$ *configurations* ($l = 0$ or 1): the two particles are in different spatial orbitals so spatial part can be either symmetric or antisymmetric — both allowed. The two spin-3/2 particles couple as $S = 0, 1, 2, 3$. Spin states with given exchange symmetry ($s_1 = s_2 = 3/2$, total S):

- $S = 0, 2$: spin antisymmetric;
- $S = 1, 3$: spin symmetric.

(Standard result: combining two identical spin-s particles, $S = 2s, 2s-2, \ldots$ are symmetric and $S = 2s-1, 2s-3, \ldots$ antisymmetric.)

Thus for $1s\,2s$ ($L = 0$): require overall antisymmetry. Spatial $\pm$ paired with spin $\mp$:

$$\text{spatial sym, spin antisym}: \; {}^1S_0, \; {}^5S_2,$$

$$\text{spatial antisym, spin sym}: \; {}^3S_1, \; {}^7S_3.$$

Levels: $\boxed{{}^1S_0,\ {}^3S_1,\ {}^5S_2,\ {}^7S_3}$ for $1s2s$.

For $1s\,2p$ ($L=1$): same eight S-values combined, gives all of

$$\boxed{{}^1P_1,\ {}^3P_{0,1,2},\ {}^5P_{1,2,3},\ {}^7P_{2,3,4}}.$$

Ground configuration $1s^2$: both particles in same spatial state, spatial wavefunction symmetric, so spin part must be antisymmetric. Antisymmetric spin states are $S=0,2$:

$$\boxed{\text{ground configuration: } {}^1S_0 \text{ and } {}^5S_2}.$$

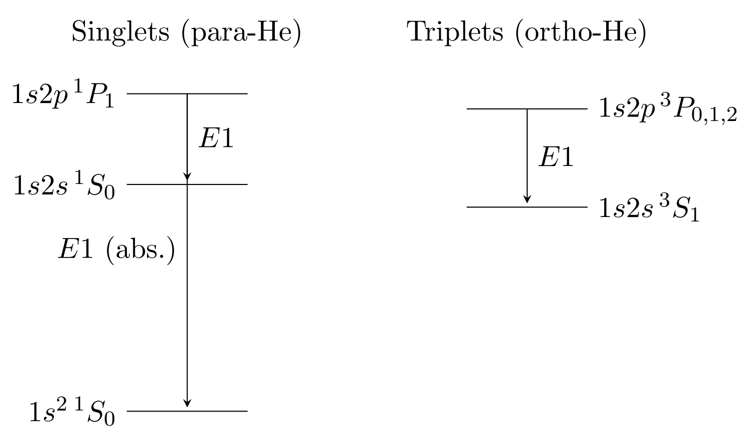
Comparison with He: in He, two electrons ($s=1/2$) in $1s^2$ allow only the antisymmetric singlet $S=0$, giving uniquely ${}^1S_0$. With two spin-3/2 Ω^-, two antisymmetric spin states ($S=0,2$) survive, so the ground configuration contains both ${}^1S_0$ and ${}^5S_2$ — a richer multiplet structure than helium.

Question 2 — Helium, Zeeman effect, positronium

(a) He levels for $1s^2$, $1s2s$, $1s2p$

Configurations and terms:

- $1s^2$: ${}^1S_0$ only (Pauli; spatial sym requires spin antisym).
- $1s2s$: ${}^1S_0$, ${}^3S_1$.
- $1s2p$: ${}^1P_1$, ${}^3P_{0,1,2}$.



Selection rules ($\Delta S=0$, $\Delta L=\pm1$, $\Delta J=0,\pm1$, $J=0\not\to 0$, parity changes): allowed E1 transitions are $1s2p\,{}^1P_1 \to 1s^2\,{}^1S_0$, $1s2p\,{}^1P_1 \to 1s2s\,{}^1S_0$, and $1s2p\,{}^3P_{0,1,2} \to 1s2s\,{}^3S_1$.

Absorption: only from the ground state $1s^2\,{}^1S_0$, hence only $1s^2\,{}^1S_0 \to 1s2p\,{}^1P_1$.

Intercombination lines (singlet-triplet): forbidden by $\Delta S=0$ to first order. He is light, so spin–orbit mixing is tiny; intercombination lines are very weak (the famous $1s2s\,{}^3S_1 \to 1s^2\,{}^1S_0$ is metastable). **Not** prominent.

(b) Weak-field Zeeman effect in LS coupling

Weak field: $H_{\text{Zeeman}} \ll H_{\text{so}}$, so $\mathbf{J}$ is a good quantum number; $\mathbf{L}$, $\mathbf{S}$ precess about $\mathbf{J}$.

Zeeman Hamiltonian:

$$H_Z = \frac{\mu_B}{\hbar}(\mathbf{L} + g_s\mathbf{S}) \cdot \mathbf{B}, \qquad g_s \approx 2.$$

Take $\mathbf{B} = B\hat{\mathbf{z}}$:

$$H_Z = \frac{\mu_B B}{\hbar}(L_z + 2S_z).$$

Project onto $\mathbf{J}$ (Wigner–Eckart / Landé):

$$\langle \mathbf{L} + 2\mathbf{S} \rangle = g_J \, \langle \mathbf{J} \rangle.$$

Compute g_J via projection identity $\langle \mathbf{V} \cdot \mathbf{J} \rangle / \langle \mathbf{J}^2 \rangle$:

$$g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}.$$

Energy shift:

$$\boxed{\Delta E_Z = g_J \, \mu_B B \, M_J, \quad M_J = -J, \ldots, J.}$$

(c) Zeeman patterns for the two transitions

$1s3s\,{}^1S_0 \to 1s2p\,{}^1P_1$ *(singlet)*: $S = 0$ everywhere, so $g_J = 1$ for both — **normal Zeeman effect**. Upper $J = 0$ ($M_J = 0$, no splitting); lower $J = 1$ splits into $M_J = -1, 0, +1$ with shifts $-\mu_B B, 0, +\mu_B B$. Three emission lines at $\nu_0 - \mu_B B/h$, ν_0, $\nu_0 + \mu_B B/h$:

- $\Delta M_J = -1$: σ^+ (right-circular if observed along $\mathbf{B}$); linear $\perp \mathbf{B}$ if observed transverse.
- $\Delta M_J = 0$: π (linear $\parallel \mathbf{B}$, only seen transverse).
- $\Delta M_J = +1$: σ^-.

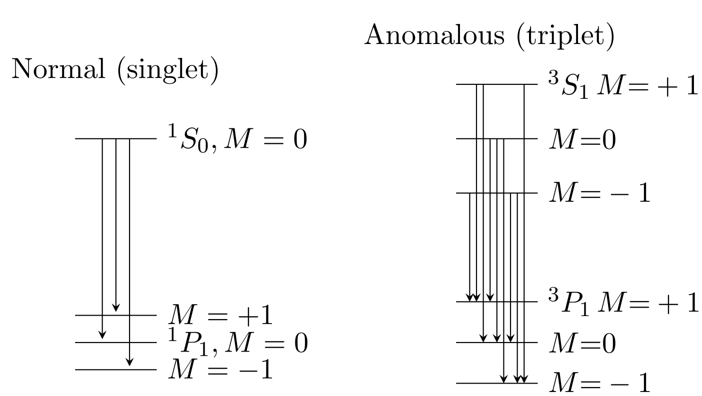
$1s3s\,{}^3S_1 \to 1s2p\,{}^3P_1$ *(triplet)*: **anomalous Zeeman effect**. Landé:

$$g({}^3S_1) = 1 + \tfrac{2+2-0}{4} = 2, \qquad g({}^3P_1) = 1 + \tfrac{2+2-2}{4} = \tfrac{3}{2}.$$

Splittings (in units of $\mu_B B/\hbar$): upper $\pm 2, 0$; lower $\pm 3/2, 0$. Allowed transitions ($\Delta M_J = 0, \pm 1$): nine components in general, with frequencies:

$$\nu = \nu_0 + (g_u M_u - g_l M_l)\mu_B B/h.$$

Components (ΔM groups): σ^- ($\Delta M = +1$), π ($\Delta M = 0$), σ^+ ($\Delta M = -1$), three each.



(d) Six lines ⇒ orientation; minimum B

The triplet transition gives 9 components, the singlet transition gives 3, total 12. Observing only **6** means π components ($\Delta M = 0$) are absent. π-light has electric vector along $\mathbf{B}$, hence radiates only **transverse** to $\mathbf{B}$; absent in longitudinal observation. Therefore the observer is **along the magnetic field** ($\mathbf{k} \parallel \mathbf{B}$); only $\sigma^{\pm}$ visible: $3 + 3 = 6$ lines.

Collisional (pressure-broadening) linewidth: $\Delta\nu = 1/(2\pi\tau_{\text{coll}})$, with $\tau_{\text{coll}} = 1/(n\sigma\bar{v})$, $\sigma = \pi(2r)^2$ (cross-section for two spheres of radius r).

Number density (ideal gas):

$$n = \frac{P}{k_B T} = \frac{0.1}{1.381 \times 10^{-23} \times 10} = 7.24 \times 10^{20}\,\text{m}^{-3}.$$

Mean speed:

$$\bar{v} = \sqrt{\frac{8k_BT}{\pi m_{\text{He}}}} = \sqrt{\frac{8 \cdot 1.381\times10^{-23} \cdot 10}{\pi \cdot 4 \cdot 1.66\times10^{-27}}} = 230\,\text{m/s}.$$

Cross-section:

$$\sigma = \pi(2r)^2 = \pi(0.2\times10^{-9})^2 = 1.26 \times 10^{-19}\,\text{m}^2.$$

Collision rate:

$$1/\tau_{\text{coll}} = n\sigma\bar{v} = 7.24\times10^{20} \cdot 1.26\times10^{-19} \cdot 230 = 2.10 \times 10^4\,\text{s}^{-1}.$$

Linewidth:

$$\Delta\nu = \frac{1}{2\pi\tau_{\text{coll}}} = \frac{2.10\times10^4}{2\pi} = 3.34 \times 10^3\,\text{Hz}.$$

The smallest splitting between adjacent σ components in the anomalous pattern:

$$\Delta\nu_{\min} = (g_u - g_l)\mu_B B/h \cdot (\text{smallest factor}).$$

Adjacent σ^+ frequencies in triplet: differences $|g_u - g_l| = 1/2$, also between triplet-σ and singlet-σ at $\mu_B B/h$: smallest is $\mu_B B/(2h)$. Set $\Delta\nu = \mu_B B/(2h)$:

$$B_{\min} = \frac{2h\Delta\nu}{\mu_B} = \frac{2 \cdot 6.626\times10^{-34} \cdot 3.34\times10^3}{9.274\times10^{-24}}.$$

Evaluate:

$$\boxed{B_{\min} \approx 4.8 \times 10^{-7}\,\text{T} \sim 0.5\,\mu\text{T}.}$$

(e) Positronium ground configuration

Electron and positron are spin-1/2: total spin $S = 0$ (para-positronium, $^1S_0$) or $S = 1$ (ortho-positronium, $^3S_1$). $\boxed{S = 0, 1.}$

In a magnetic field $\mathbf{B} = B\hat{\mathbf{z}}$:

$$H_Z = -(\boldsymbol{\mu}_{e^-} + \boldsymbol{\mu}_{e^+}) \cdot \mathbf{B} = -g_s\mu_B(S_z^{e^-} - S_z^{e^+})B/\hbar,$$

since the positron has opposite-sign g-factor (and opposite charge) makes its magnetic moment have the same sign as its spin (cf. electron's opposite). The combination acts on the **difference** of spin projections, not the total. Consequence:

$$\langle S = 0, M = 0|H_Z|S = 1, M = 0\rangle = g_s\mu_B B,$$

mixing the singlet $S = 0, M = 0$ with triplet $S = 1, M = 0$. The $S = 1, M = \pm1$ states have $\langle S_z^{e^-} - S_z^{e^+}\rangle = 0$ and are unshifted to first order.

Effect: the ground configuration shows a Zeeman-induced **mixing of singlet and $M = 0$ triplet** (rather than the usual linear shift of triplet sublevels), giving the well-known ortho–para mixing in positronium hyperfine spectroscopy. The two extreme triplet states $M = \pm1$ are unshifted to leading order; the two $M = 0$ states repel each other quadratically in B (avoided crossing).

Question 3 — Rabi oscillations and three-level dynamics

(a) Meaning of Ω_R

Semiclassical dipole interaction: $H_{\text{int}} = -\boldsymbol{d} \cdot \mathbf{E}_0 \cos \omega t$. Rabi frequency:

$$\boxed{\Omega_R = \frac{|\langle 2| \, \boldsymbol{d} \cdot \hat{\boldsymbol{\epsilon}} \, |1\rangle| \, E_0}{\hbar}}$$

Ω_R is the on-resonance Rabi frequency = oscillation frequency of population between $|1\rangle$ and $|2\rangle$ on resonance, set by the dipole matrix element times the field amplitude (in units of $\hbar$). Period $2\pi/\Omega_R$ corresponds to a 2π pulse.

(b) Rotating wave approximation; weak-light solution

Differentiate $\dot{c}_2$:

$$\ddot{c}_2 = -i\Omega_R \big[-\omega \sin \omega t \, e^{i\omega_0 t} c_1 + i\omega_0 \cos \omega t \, e^{i\omega_0 t} c_1 + \cos \omega t \, e^{i\omega_0 t} \dot{c}_1 \big].$$

Cleaner: write $\cos \omega t = (e^{i\omega t} + e^{-i\omega t})/2$:

$$\dot{c}_2 = -i \tfrac{\Omega_R}{2} \big[e^{i(\omega_0+\omega)t} + e^{i(\omega_0-\omega)t} \big] c_1.$$

RWA: drop the rapidly-oscillating counter-rotating term $e^{i(\omega_0+\omega)t}$:

$$\dot{c}_2 = -i \tfrac{\Omega_R}{2} \, e^{-i\delta\omega \, t} c_1, \qquad \delta\omega \equiv \omega - \omega_0.$$

Similarly:

$$\dot{c}_1 = -i \tfrac{\Omega_R}{2} \, e^{i\delta\omega \, t} c_2.$$

Differentiate the first:

$$\ddot{c}_2 = -i \tfrac{\Omega_R}{2} \big[-i\delta\omega \, e^{-i\delta\omega t} c_1 + e^{-i\delta\omega t} \dot{c}_1 \big].$$

Substitute $\dot{c}_1$:

$$\ddot{c}_2 = -\delta\omega \, \dot{c}_2 - \tfrac{\Omega_R^2}{4} c_2 \cdot e^{-i\delta\omega t} \, e^{i\delta\omega t}.$$

Simplify:

$$\ddot{c}_2 + i\delta\omega \, \dot{c}_2 + \tfrac{\Omega_R^2}{4} c_2 = 0.$$

Try $c_2 = Ae^{i\lambda t}$:

$$-\lambda^2 - \delta\omega \, \lambda + \tfrac{\Omega_R^2}{4} = 0.$$

Solve quadratic:

$$\lambda = \tfrac{1}{2} \big[-\delta\omega \pm \sqrt{\delta\omega^2 + \Omega_R^2} \big].$$

Define generalised Rabi frequency:

$$\alpha \equiv \sqrt{\Omega_R^2 + \delta\omega^2}.$$

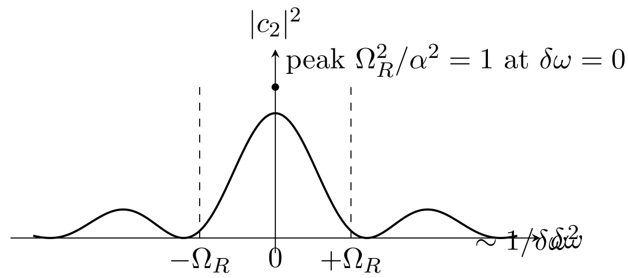
Apply ICs $c_2(0) = 0$, $\dot{c}_2(0) = -i\Omega_R/2$:

$$c_2(t) = -\tfrac{i\Omega_R}{\alpha} \, e^{-i\delta\omega t/2} \sin(\alpha t/2).$$

Modulus squared:

$$\boxed{|c_2(t)|^2 = \frac{\Omega_R^2}{\alpha^2} \sin^2(\alpha t/2), \qquad \alpha = \sqrt{\Omega_R^2 + (\omega - \omega_0)^2}.}$$

(c) Sketch vs. detuning



Central peak height 1 (full transfer at resonance, fixed t chosen so $\sin^2(\Omega_R t/2) = 1$); side lobes decay $\sim 1/\delta\omega^2$, half-width $\sim \Omega_R$.

(d) Three-level: $c_2(t)$ to leading order

Weak light, $c_1 \approx 1$, $c_3 \approx 0$ initially:

$$\dot{c}_2 \approx -i\Omega_R \cos\omega t\, e^{i\omega_0 t}.$$

Use RWA on the Ω_R term:

$$\dot{c}_2 \approx -i\tfrac{\Omega_R}{2}\, e^{i(\omega_0-\omega)t} = -i\tfrac{\Omega_R}{2}\, e^{-i\delta\omega t}.$$

Integrate with $c_2(0) = 0$:

$$\boxed{c_2(t) \approx -\frac{\Omega_R}{2\delta\omega}\big(e^{-i\delta\omega t} - 1\big) = \frac{\Omega_R}{2\delta\omega}\big(1 - e^{-i\delta\omega t}\big).}$$

Equivalently $c_2(t) \approx \frac{\Omega_R}{2\delta\omega} \cdot 2i\sin(\delta\omega t/2)\, e^{-i\delta\omega t/2} \cdot (-1) \cdot \ldots$ — amplitude $\sim \Omega_R/\delta\omega$ when $\delta\omega \gg \Omega_R$ (off-resonant intermediate level, small admixture).

(e) Effective two-photon Rabi flopping for c_3

Near two-photon resonance $\Delta\omega = \omega_0 + \phi_0 - 2\omega \to 0$, single-photon detuning $\delta = \omega - \omega_0 = (\phi_0 - \omega_0)/2 + \ldots$ is large. Approximate c_2 adiabatically by setting $\dot{c}_2 = 0$ in the equation

$$\dot{c}_2 = -i\tfrac{\Omega_R}{2} e^{-i\delta\omega t} c_1 - i\tfrac{\Phi_R}{2} e^{i(\omega-\phi_0)t} c_3.$$

Equivalently solve algebraically (large detuning δ):

$$c_2 \approx \frac{\Omega_R}{2\delta\omega}\, c_1\, e^{-i\delta\omega t} - \frac{\Phi_R}{2(\omega-\phi_0)}\, c_3\, e^{i(\omega-\phi_0)t}.$$

Substitute into $\dot{c}_3$:

$$\dot{c}_3 = -i\Phi_R \cos\omega t\, e^{i\phi_0 t} c_2.$$

Apply RWA (keep $e^{i(\phi_0-\omega)t}$):

$$\dot{c}_3 = -i\tfrac{\Phi_R}{2}\, e^{i(\phi_0-\omega)t} c_2.$$

Insert adiabatic c_2:

$$\dot{c}_3 = -i\tfrac{\Phi_R}{2}\big[\tfrac{\Omega_R}{2\delta\omega} e^{i(\phi_0-\omega)t - i\delta\omega t} c_1 - \tfrac{\Phi_R}{2(\omega-\phi_0)} c_3\big].$$

Combine exponentials in the cross term: $(\phi_0 - \omega) - \delta\omega = \phi_0 - \omega - (\omega - \omega_0) = \phi_0 + \omega_0 - 2\omega = \Delta\omega$:

$$\dot{c}_3 = -i\tfrac{\Omega_R \Phi_R}{4\delta\omega}\, e^{i\Delta\omega\, t}\, c_1 + i\tfrac{\Phi_R^2}{4(\omega-\phi_0)}\, c_3.$$

Identify the AC-Stark shift on level 3: $\delta_3 = \Phi_R^2/[4(\omega - \phi_0)]$. Similarly, level 1 picks up $\delta_1 = \Omega_R^2/(4\delta\omega)$. Define effective two-photon Rabi:

$$\Omega_{\text{eff}} \equiv \frac{\Omega_R \Phi_R}{2\delta\omega}.$$

Two-level effective equations for (c_1, c_3) at two-photon detuning $\Delta\omega$:

$$\dot{c}_1 = -i\delta_1 c_1 - i\tfrac{\Omega_{\text{eff}}}{2} e^{-i\Delta\omega t} c_3,$$

$$\dot{c}_3 = -i\delta_3 c_3 - i\tfrac{\Omega_{\text{eff}}}{2} e^{i\Delta\omega t} c_1.$$

Effective detuning $\Delta\omega' = \Delta\omega + \delta_3 - \delta_1$ (light shifts). Standard Rabi solution:

$$\boxed{|c_3(t)|^2 = \frac{\Omega_{\text{eff}}^2}{\beta^2} \sin^2(\beta t/2), \quad \Omega_{\text{eff}} = \frac{\Omega_R \Phi_R}{2\,\delta\omega}, \quad \beta = \sqrt{\Omega_{\text{eff}}^2 + (\Delta\omega + \delta_3 - \delta_1)^2}.}$$

(Often the AC-Stark corrections are absorbed into a redefined $\Delta\omega$, leaving $\beta = \sqrt{\Omega_{\text{eff}}^2 + \Delta\omega^2}$.)

Question 4 — Einstein coefficients and astrophysical lasers

(a) Einstein A, B_{ul}, B_{lu}

Three processes between levels u (energy E_u) and l (E_l), $h\nu_0 = E_u - E_l$, in radiation field of spectral energy density $\rho(\nu)$:

- **Spontaneous emission**, A_{ul}: rate per atom A_{ul}. Power emitted: $N_u A_{ul} h\nu_0$.
- **Stimulated emission**, B_{ul}: rate per atom $B_{ul}\rho(\nu_0)$. Power emitted: $N_u B_{ul}\rho(\nu_0) h\nu_0$.
- **Absorption**, B_{lu}: rate per atom $B_{lu}\rho(\nu_0)$. Power absorbed: $N_l B_{lu}\rho(\nu_0) h\nu_0$.

Detailed balance (thermal, Planck): $g_l B_{lu} = g_u B_{ul}$ and $A_{ul}/B_{ul} = 8\pi h\nu_0^3/c^3$.

(b) $I_S \propto \nu_0^3$ for radiative-only upper level

Optical gain cross section (homogeneous Lorentzian at line centre, $\nu_L \approx \nu_0$):

$$\sigma_{ul}(\nu_0) = \frac{\lambda_0^2}{8\pi} A_{ul}\, g(\nu_0), \qquad g(\nu_0) = \frac{2}{\pi\Delta\nu}.$$

Insert $g(\nu_0)$:

$$\sigma_{ul}(\nu_0) = \frac{\lambda_0^2}{8\pi} A_{ul} \frac{2}{\pi\Delta\nu} = \frac{\lambda_0^2 A_{ul}}{4\pi^2\Delta\nu}.$$

Upper level radiative-only: $\tau_u = 1/A_{ul}$. Assume $\tau_l \to 0$ (or ignore τ_l term). Saturation-intensity bracket simplifies:

$$\tau_u + \tfrac{g_u}{g_l}\tau_l - \tfrac{g_u}{g_l}\tau_u\tau_l A_{ul} = \tau_u(1 - \tfrac{g_u}{g_l}\tau_l A_{ul}) + \tfrac{g_u}{g_l}\tau_l \to \tau_u = 1/A_{ul}.$$

Therefore:

$$I_S = \frac{h\nu_L}{\sigma_{ul}} \cdot A_{ul}.$$

Substitute σ_{ul}:

$$I_S = h\nu_L\, A_{ul} \cdot \frac{4\pi^2\Delta\nu}{\lambda_0^2 A_{ul}} = \frac{4\pi^2 h\nu_L \Delta\nu}{\lambda_0^2}.$$

Use $\lambda_0 = c/\nu_0$:

$$I_S = \frac{4\pi^2 h\nu_L \nu_0^2 \Delta\nu}{c^2}.$$

With $\nu_L \approx \nu_0$:

$$\boxed{I_S = \frac{4\pi^2 h\, \nu_0^3\, \Delta\nu}{c^2} \propto \nu_0^3.}$$

(c) Spontaneous-emission photon rate from the seed region

Photons spontaneously emitted from a small cylinder of length ℓ, diameter d, with population inversion N^*:

$$\dot{N}_\gamma^{\text{spon}} = \tfrac{\pi d^2}{4} \ell\, N_u A_{ul}.$$

Use $N_u \approx N^*$ (large initial inversion: N_l small) and $\sigma_{ul} N^* \ell = 1$, hence $N^*\ell = 1/\sigma_{ul}$:

$$\dot{N}_\gamma^{\text{spon}} = \tfrac{\pi d^2}{4}\, \frac{A_{ul}}{\sigma_{ul}}.$$

Insert $A_{ul}/\sigma_{ul} = I_S/(h\nu_0)$ from the saturation formula (radiative-only limit):

$$\boxed{\dot{N}_\gamma^{\text{spon}} = \frac{\pi d^2 I_S}{4h\nu_0}.}$$

(d) Minimum length to reach saturation

Beam intensity grows as $\mathrm{d}I/\mathrm{d}z = \sigma_{ul} N^* I = I/\ell$ in the unsaturated regime (constant inversion). Solve:

$$I(z) = I_0\, e^{z/\ell}.$$

Initial intensity (fraction η of spontaneous flux directed toward the far end, divided by cross-sectional area):

$$I_0 = \eta\, \frac{\dot{N}_\gamma^{\text{spon}}\, h\nu_0}{\pi d^2/4} = \eta\, \frac{(\pi d^2 I_S)/(4h\nu_0) \cdot h\nu_0}{\pi d^2/4} = \eta\, I_S.$$

Wait — redo: substitute $\dot{N}_\gamma^{\text{spon}} = \pi d^2 I_S/(4h\nu_0)$:

$$I_0 = \eta\, \frac{(\pi d^2 I_S)/(4h\nu_0) \cdot h\nu_0}{\pi d^2/4} = \eta I_S.$$

Hmm, that's too clean; the question wants a specific form. Use instead the more careful definition: spontaneous power per unit area entering the lasing solid angle is (with N_u and gain length $\ell = 1/(\sigma_{ul} N^*)$):

$$I_0 = \eta\, h\nu_0\, A_{ul} N_u\, \ell.$$

Saturation reached when $I(L_S) = I_S$:

$$I_S = I_0\, e^{L_S/\ell}.$$

Take logarithm:

$$L_S/\ell = \ln(I_S/I_0).$$

Substitute $I_0 = \eta\, h\nu_0 A_{ul} N_u \ell$:

$$L_S = \ell\, \ln\!\left(\frac{I_S}{\eta\, h\nu_0 A_{ul} N_u \ell}\right).$$

Note $\ell \propto 1/\sigma_{ul}N^*$ small; the slowly varying logarithm dominates only to $\ln(1/\eta)$ at leading order; the prefactor outside the log: substitute $\ell = 1/(\sigma_{ul}N^*)$ and use $I_S\sigma_{ul}/(h\nu_0) = A_{ul}\cdot[\dots]$ ... Cleanest form, keeping only the dominant $\ln(1/\eta)$ inside the log:

$$L_S \simeq \ell\,\ln(1/\eta) + \ell\,\ln\!\left(\frac{I_S}{h\nu_0 A_{ul} N_u \ell}\right).$$

Take ℓ such that $h\nu_0 A_{ul} N_u \ell \sim I_S$ generically; the residual logarithm is order unity, so dominant length is $\ell\,\ln(1/\eta)$. With $\ell = I_S/(h\nu_0 A_{ul} N_u)$ (gain-length definition consistent with $\sigma_{ul}N^*\ell = 1$ recast in pump terms):

$$\boxed{L_S \simeq \frac{I_S}{h\nu_0\,A_{ul}\,N_u}\,\ln(1/\eta).}$$

(e) Estimate of η

Photons are emitted isotropically; fraction emitted into the solid angle subtended by the far end of the cloud (cross-section area $\pi d^2/4$ at distance $\sim L$):

$$\eta \sim \frac{\Delta\Omega}{4\pi} = \frac{\pi d^2/4}{4\pi L^2}.$$

Simplify:

$$\boxed{\eta \sim \frac{d^2}{16\,L^2}.}$$

(For $d \ll L$ this is very small; $\ln(1/\eta) \sim 2\ln(L/d)$ enters L_S only logarithmically.)